

Motivation for cloud computing

cloud computing is driven by ~~sent~~ several key motivations that make it appealing for businesses, developers and individuals.

cost efficiency:-

eliminates upfront hardware and infrastructure costs. Pay-as-you-go models allow scaling resources to demand, reducing waste.

scalability:-

easily scale resources up or down to handle varying workloads, ensuring performance without over-provisioning.

flexibility and accessibility:-

Access data and applications from anywhere with an internet connection, supporting remote work and global collaboration.

Speed and agility:-

Rapid deployment of applications and services, enabling faster innovation and time-to-market for businesses.

Security:-

Advanced security features, regular updates, and compliance with industry standards protect data and applications.

Maintenance-free:-

Provide handle hardware maintenance, software updates and infrastructure management freeing up IT teams for strategic tasks.

collaboration and Productivity:-

cloud based tools enable real-time collaboration, streamlining workflows and boosting efficiency.

Environmental impact:-

shared resources and optimized data centers reduce energy consumption compared to on-premises setup.

innovation enablement:-

Access to cutting edge technologies like AI, machine learning, and big data analytics without significant upfront investment

* Five essential characteristics:-

There, five characteristics:-

(1) on-demand self-services:-

The cloud computing services does not require any human administrators, user themselves are able to provision, ~~monitor~~ monitor and manage computing resources as needed.

(2) ~~on-demand~~

(2) Rapid elasticity:-

The computing services should have IT resources that are able to scale out and in quickly and on a need basis.

* Whenever the user require services it is provided to him and it is scale out

as soon as its requirement gets over.

③ Broad network access:-

The computing services^{are} generally provided over standard networks and heterogeneous devices. (systems that use more than one kind of processor or core).

④ Elasticity Resource Pooling:-

The IT resource (eg. networks, servers, storage, applications and services) present and shared across multiple applications and occupant in the uncommitted manner. multiple clients are provided service from a same physical resource.

⑤ Measured service:- occupant's user who has access to and utilizes resources within a shared cloud environment.

The resource utilization is tracked for each application and occupant, it will provide both the use and the resource provided with an account of what has been used. This is done for various reasons like monitoring billing and effective use of resource.

Need for cloud computing:-

① Growing Data and storage Requirement

(i) organizations generates vast amount of data daily.

(ii) cloud offers virtually unlimited storage capacity without needing physical hardware.

② High Infrastructure and maintenance costs

(i) setting up and maintaining physical servers is expensive.

~~Cloud reduces capital capex~~

(ii) cloud reduces capital and operational expenses with a pay-as-you-use model.

③ Scalability and Flexibility:-

(i) Businesses experience fluctuating workloads.

(ii) cloud allows easy scaling of resources up or down as needed.

④ Global accessibility and mobility:-

(i) With remote work and global teams, access to resources from anywhere is crucial.

(ii) cloud enables 24/7 access via internet

⑤ Faster Deployment and Faster development:-

(i) cloud platforms offer tools and environments to quickly build and deploy applications.

(ii) speeds up innovation and time-to-market.

⑥ Improved collaboration:-

(i) Teams can collaborate in real-time using cloud based tools.

(ii) shared access to files, apps, and data improves productivity.

⑦ Disaster Recovery and Backup:-

(i) Local hardware failures can cause data loss.

(ii) cloud provides automatic backups and disaster recovery options.

⑧ security and compliance:-

(i) cloud Providers invest heavily in security infrastructure.

(ii) Many offer compliance with global standards like ISO, GDPR, HIPAA etc.

(9) Need for Green IT:-
General data protection Regulation
Health Insurance Portability and Accountability Act

(i) cloud computing reduces energy usage through efficient resource sharing

(ii) ~~also~~ supports eco-friendly computing.

⑩ supports for emerging technologies:-

(i) cloud supports AI, machine learning, big data, and IoT.

(ii) These technologies need high processing power and storage, which cloud offers on-demand.